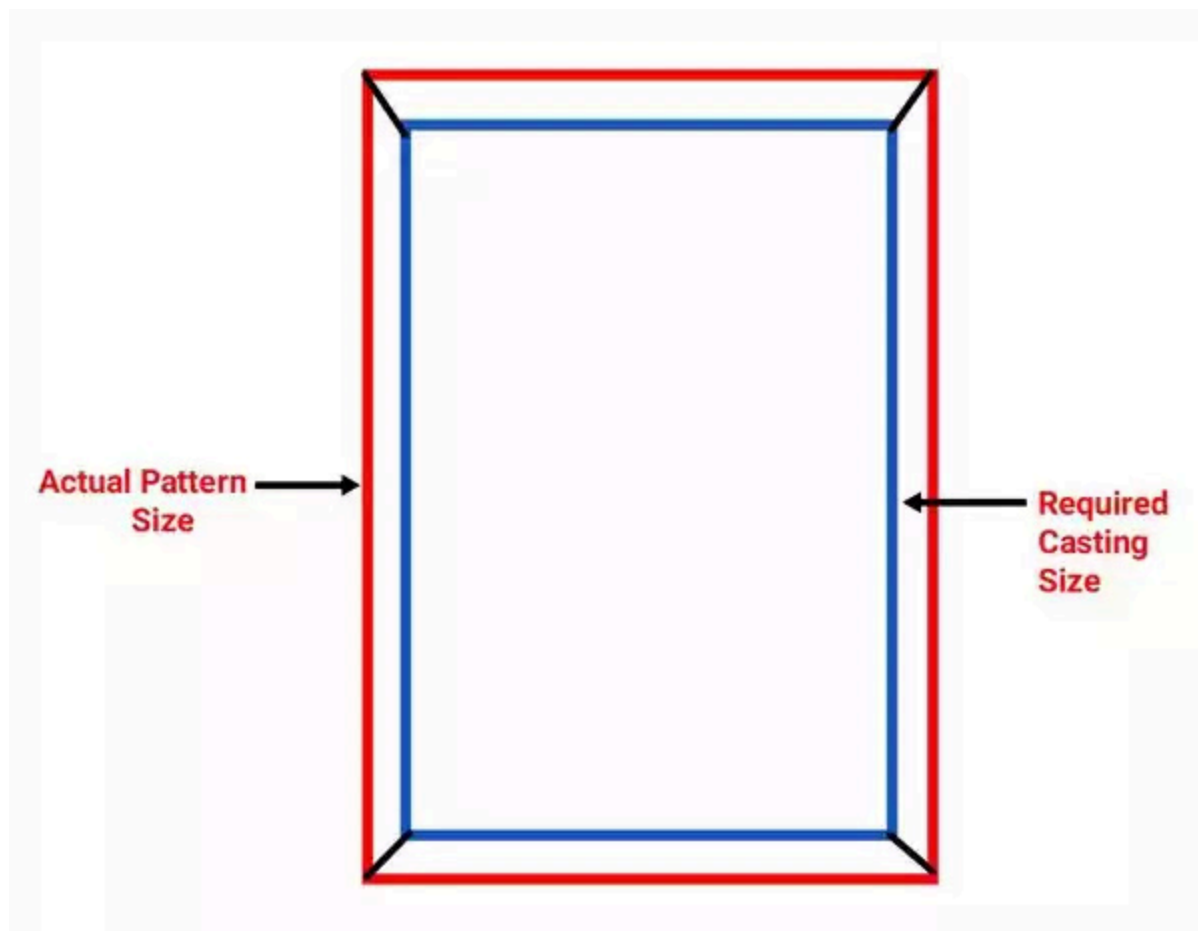


Question 9

Types of allowances and their sketches

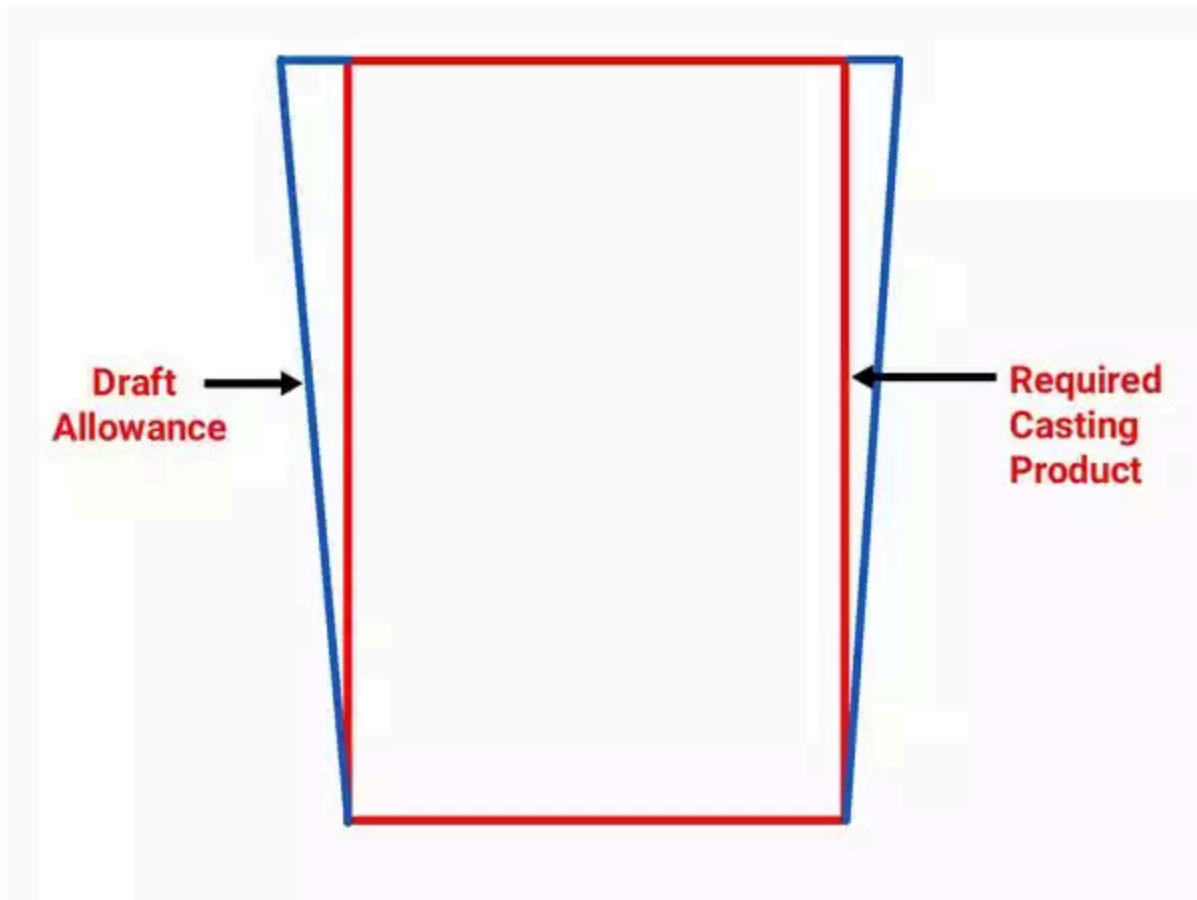
Shrinkage or Contraction Process

1. Liquid shrinkage - To account for the reduction in volume when the metal changes from liquid state to solid state at the solidus temperature.
2. Solid shrinkage - To account for the reduction in volume caused when metal loses temperature in solid state.



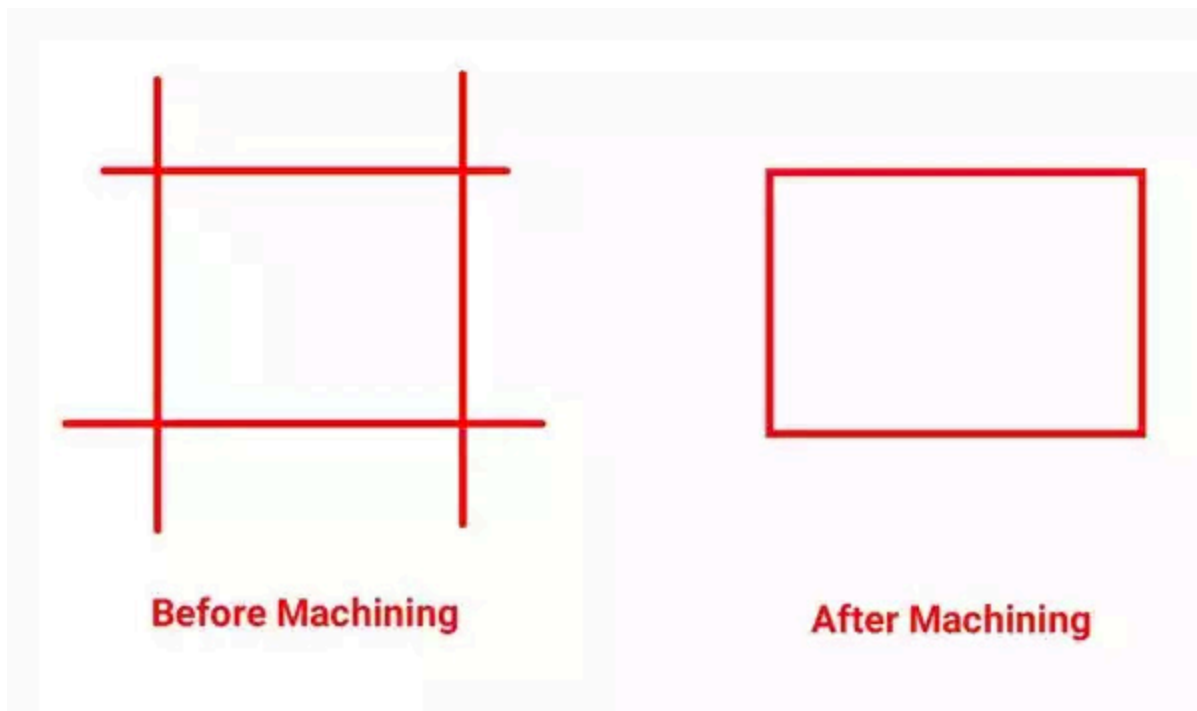
Draft or Taper Allowance

The taper provided by the pattern maker on all vertical surfaces of the pattern so that it can be removed from the sand without tearing away the sides of the sand mold and without excessive rapping by the molder.



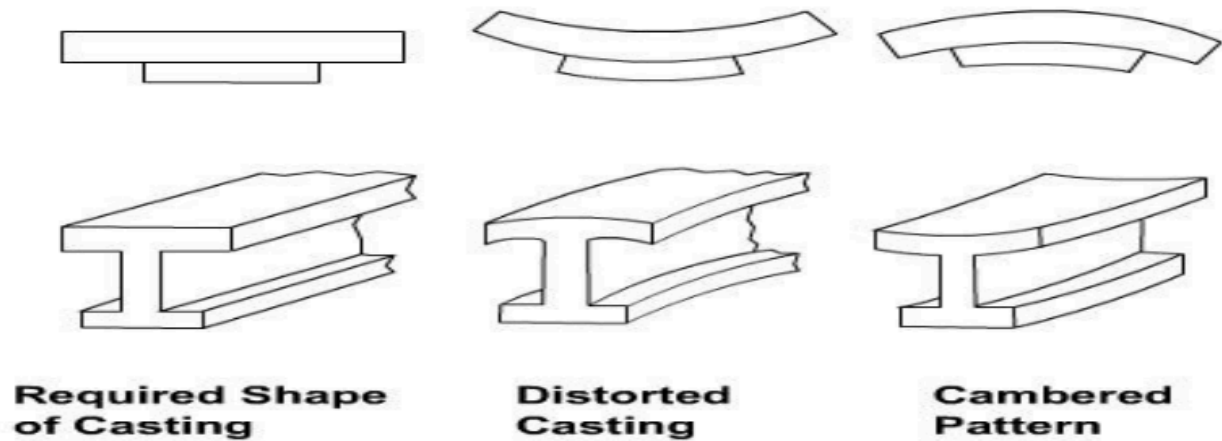
Machining or Finishing Allowance

Allowances added for subsequent machining because of the sand casting process' poor surface finish and dimensional accuracy.



Distortion or Camber Allowance

Adjustments made to counteract warping or bending during cooling. It's common for irregular shapes.



Rapping Allowance

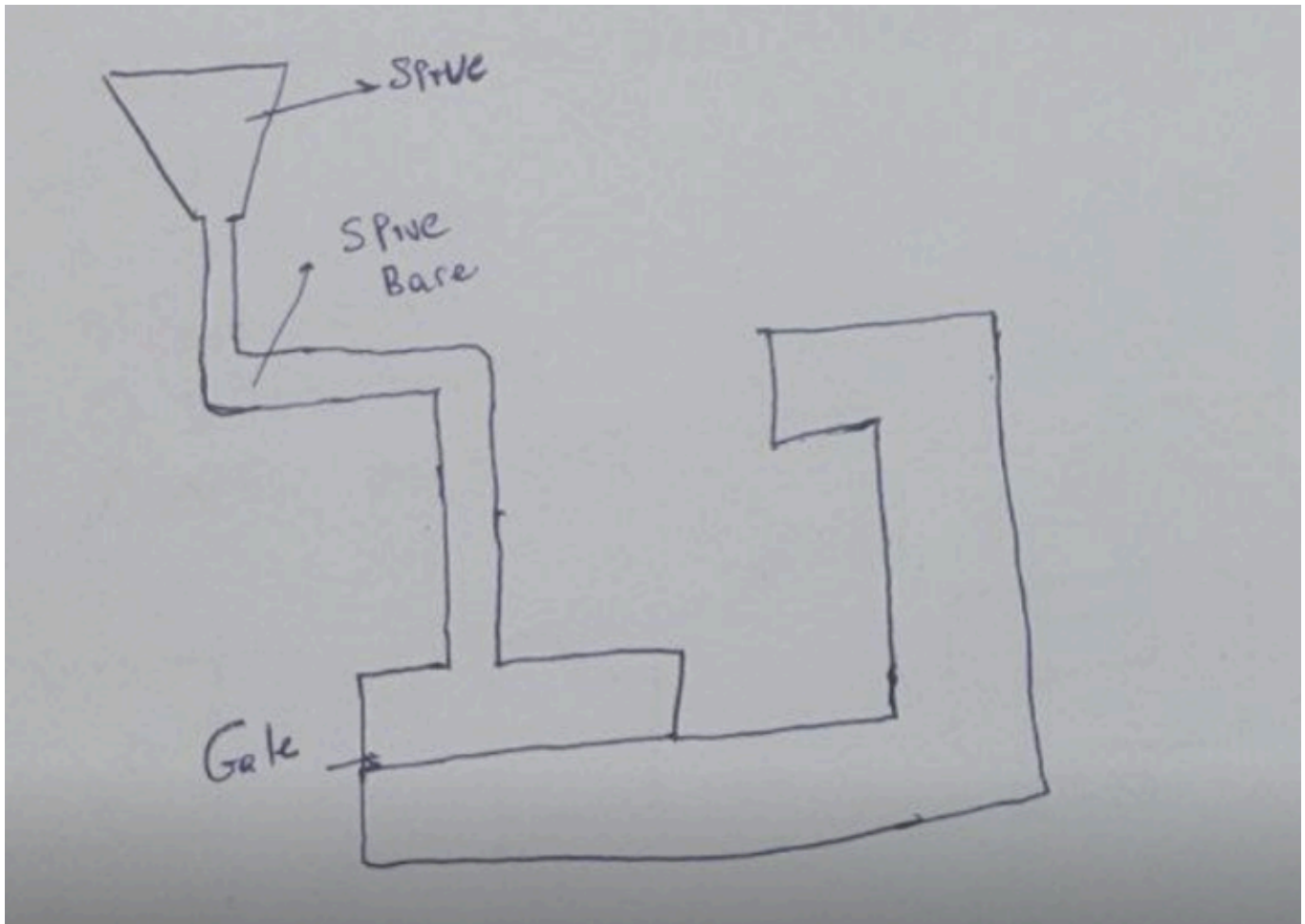
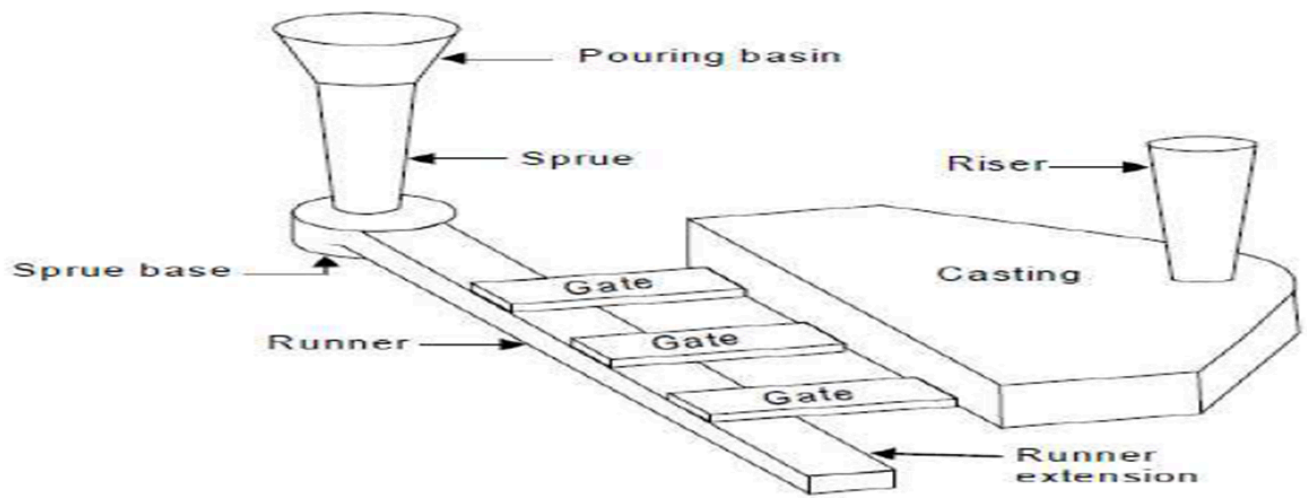
Compensates for the enlargement of the mould cavity due to rapping (tapping) the pattern before removal. It is a negative allowance and is to be applied only to those dimensions that are parallel to the parting plane.

What are the requirements of a good gated system?

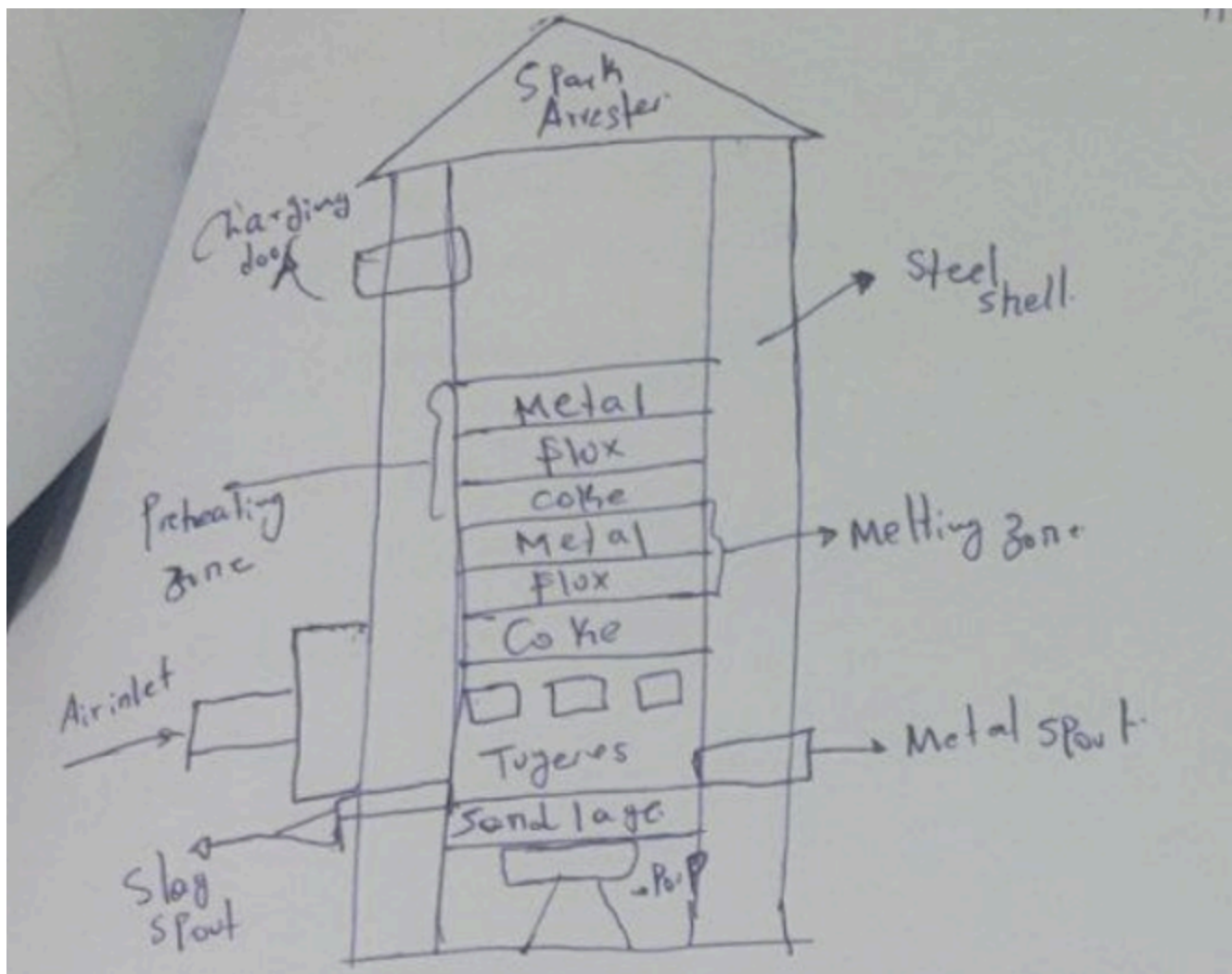
A **gating system** ensures that molten metal enters the mould cavity effectively. The requirements are:

1. **Smooth and controlled flow** of molten metal.
2. **Minimum turbulence** to avoid air entrapment and oxidation.
3. **Proper directional solidification**, aided by risers.
4. **Efficient removal of slag and dross** using a well-designed pouring basin.
5. **Easy and economical removal** of gating material after casting.
6. **Avoid mould erosion** by reducing metal velocity.
7. Maintain **temperature of molten metal** during flow.

Sketch of gated system



Question 10



Operation

1. Preparation/repair of refractory lining
2. Lighting & burning in the coke bed
3. Charging
4. Melting:
 - starting the air blast
 - charging
5. Tapping and slagging
6. Dropping the bottom

Construction

- **Charging Door** – At the top, used to add metal, coke, and flux.
- **Stack/Chimney** – Releases hot gases and smoke.
- **Shell** – Steel body of the furnace.
- **Refractory Lining** – Heat-resistant inner lining.

- **Tuyeres** – Openings near the bottom to supply air for combustion.
- **Wind Box & Blower** – Delivers air to tuyeres.
- **Crucible (Well)** – Collects molten metal.
- **Slag Hole** – Removes slag (impurities).
- **Tap Hole** – Drains molten metal.
- **Bottom Door** – Opens to discharge residue after use.

Question 11

A) Design Consideration for Castability

- Uniform wall thickness
- Proper draft angles

B) Avoiding Defects

- Proper gating system design
- Adequate riser placement

C) Economic and Manufacturing Factors

- Minimize machining

Question 12

Works on **electromagnetic induction**.

An alternating current in a coil creates a magnetic field, which induces **eddy currents** in the metal. These currents generate **heat** due to resistance, melting the metal.

- Metal is placed in a crucible inside a coil.
- AC flows through the coil, generating a magnetic field.
- Eddy currents heat and melt the metal.
- Molten metal is poured after reaching the desired temperature.